Software Requirements Specification

for

Daddict: Side-Scroller Edition

Version 1.0 approved

Date Created: 28/02/2025

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Table of Contents

1. Introduction

This document outlines the requirements for "Daddict: Side-Scroller Edition", a 2.5D side-scrolling game where players navigate through a dynamic environment filled with enemies and drugs. Players must manage their health by avoiding drugs that reduce their life while collecting medicines to restore health.

1.1 Document Purpose

This document specifies the software requirements for "Daddict: Side-Scroller Edition," version 1.0. It is intended for various stakeholders involved in the development and deployment of the game, including:

- **Developers**: To understand the technical requirements and functionalities needed for implementation.
- **Project Managers**: To oversee the project timeline, resource allocation, and ensure that the development aligns with the specified requirements.
- **Marketing Staff**: To gain insights into the game's features and unique selling points for promotional activities.
- **Users**: To provide an understanding of what to expect from the game and its mechanics.
- **Testers**: To develop test cases and ensure that the game meets the specified requirements and quality standards.
- **Documentation Writers**: To create user manuals and other supporting documentation based on the requirements outlined in this SRS.

1.2 Document Conventions

1. **Text Styles**:
    - Bold: Used to highlight key terms and important concepts within the text.
    - *Italic*: Used for emphasis and to denote titles of other documents, such as related game design documents or references.
    - Monospace: Used for code snippets, variable names, or specific technical terms that require precise formatting.
2. **Headings**:

- Headings are organized hierarchically, with primary headings numbered as 1, 2, 3, etc., and subheadings numbered as 1.1, 1.2, 1.3, etc. This structure allows for easy navigation through the document.

3. **Requirement Identifiers**:
    - Each unique requirement will be labeled with a unique identifier in the format REQ-XX, where "XX" is a sequential number starting from 01 (e.g., REQ-01, REQ-02). This identifier will be used throughout the document to reference specific requirements easily.

1.3 Project Scope

"Daddict: Side-Scroller Edition" is a 2.5D side-scrolling game designed to engage players in a dynamic and immersive experience while exploring themes of addiction and recovery. The primary purpose of the software is to provide entertainment while subtly raising awareness about the consequences of drug use and the importance of health management.

Business Objectives and Strategies

- **Engagement**: To create a captivating gameplay experience that keeps players returning, thereby increasing user retention and engagement metrics.
- **Awareness**: To promote awareness of addiction and recovery through gameplay, potentially partnering with organizations focused on addiction support and education.
- **Monetization**: To explore monetization strategies such as in-game purchases for recovery items or cosmetic upgrades, while ensuring that the game remains accessible and enjoyable for all players.

Major Features

- **Endless Running Mechanics**: Players navigate through an infinite landscape, avoiding obstacles and collecting items.
- **Drug Consumption Effects**: The game simulates the effects of drug use, including decreased health, creating a challenging experience.
- **Health Management**: Players can find medicines that help restore health, promoting the theme of recovery.
- **Dynamic Difficulty Scaling**: The game becomes progressively harder over time, increasing the challenge and engagement for players.
- **Visual and Audio Feedback**: The game employs visual effects (e.g., darkening screens, blurry vision) and audio cues to enhance the immersive experience and convey the player's status.

1.4 References

These references provide additional context, guidelines, and standards that are relevant to the development and design of the game.

1. **Game Design Document for Daddict: Side-Scroller Edition**
    - **Source:https://docs.google.com/document/d/1piL1z7spjf5zyQ1CNlx4mmUeigh6wbBVNWaKnuo4IEc/edit?usp=sharing**
2. **Unity Documentation**
    - **Source: https://docs.unity3d.com/**
3. **C# Scripting API**
    - **Source: https://learn.microsoft.com/en-us/dotnet/csharp/**
4. **Blender Modeling Tutorials**
    - **Source: https://www.blender.org/support/tutorials/**
5. **Freedom Club**
    - **Source: https://devfreedom.club/**

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2. Overall Description

This section presents a high-level overview of the product and the environment in which it will be used, the anticipated users, and known constraints, assumptions, and dependencies.

2.1 Product Perspective

"Daddict: Side-Scroller Edition" is an entirely new product that combines elements of endless running and side-scrolling gameplay with themes of addiction and recovery. It is not a replacement for an existing application nor an iteration of a previous version. The game will be developed as a standalone application, primarily targeting PC and mobile platforms.

The game will feature a 2.5D perspective, where players continuously move to the right, encountering enemies and obstacles while managing their health through drug consumption and medicine collection. The game will be designed to raise awareness about addiction while providing an engaging and entertaining experience.

2.2 User Classes and Characteristics

The anticipated user classes for "Daddict: Side-Scroller Edition" include:

- **Casual Gamers**: Players looking for a fun and engaging experience without complex mechanics. They may have varying levels of gaming experience and are primarily interested in entertainment.
- **Gamers with Awareness**: Players who understand the themes of addiction and recovery and are interested in narrative-driven gameplay. They may seek games that provide deeper messages and social commentary.
- **Health-Conscious Players**: Individuals who are interested in games that promote health awareness and education regarding addiction and recovery.
- **Developers and Testers**: Individuals involved in the development and testing of the game who require access to the game's mechanics and functionalities.

2.3 Operating Environment

The game will operate in the following environments:

- **Hardware Platforms:**
  - PCs with Windows and macOS operating systems.
  - Mobile devices running iOS and Android.
- **Software Requirements:**
  - Unity game engine for development.
  - C# for scripting.
  - Blender for 3D modeling and animations.
  - Freedom Club for assets
  - Maximo for characters
- **Geographical Locations:**
  - The game will be accessible globally, with no specific geographical restrictions.
- **Coexisting Software:**
  - The game must peacefully coexist with various operating systems and may require integration with social media platforms for sharing scores and achievements.

2.4 Design and Implementation Constraints

The following constraints will limit the options available to developers:

- **Regulatory Policies:** Compliance with age rating and content guidelines for games, particularly those addressing sensitive topics like addiction.

- **Hardware Limitations:** The game must run smoothly on devices with varying specifications, targeting a minimum of 30 frames per second.
- **Technological Constraints**: The game will be developed using Unity and C#, which may limit certain functionalities based on the capabilities of these tools.
- **Art and Asset Limitations**: The game will utilize available assets from the Unity Asset Store and other sources, which may impose restrictions on the visual style and animations.

2.5 Assumptions and Dependencies

The following assumptions and dependencies are identified:

- **Assumptions:**
    - Players have basic knowledge of how to play 2D running games.
    - The game will require internet access for updates and leaderboards.
- **Dependencies:**
    - The game relies on third-party assets and libraries from the Unity Asset Store and other sources.
    - The game's performance may depend on the hardware capabilities of the user's device.
    - The game may depend on external APIs for social sharing features and leaderboards.

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3. System Features

This section organizes the functional requirements for "Daddict: Side-Scroller Edition" by system features, detailing the major services provided by the product.

3.1 Endless Running Mechanic

3.1.1 Description: The player character continuously moves to the right side of the screen. This feature is of **High** priority.

3.1.2 Stimulus/Response Sequences:

- **Stimulus:** The game starts, and the player character begins moving automatically.
- **Response:** The player can perform actions such as jumping, dashing, or crouching.

3.1.3 Functional Requirements:

- REQ-01: The player character must move forward automatically at a constant speed.
- REQ-02: The speed of the player character must gradually increase as the game progresses.
- REQ-03: The player must be able to pause the game at any time.

3.2 Jumping Mechanic

3.2.1 Description: The player can jump over obstacles and enemies. This feature is of **High** priority.

3.2.2 Stimulus/Response Sequences:

- **Stimulus**: The player presses the jump button.
- **Response**: The player character jumps over obstacles or enemies.

3.2.3 Functional Requirements:

- REQ-04: The player must be able to jump a specified height.
- REQ-05: The jump action must be responsive and allow for variable jump heights based on how long the jump button is held.
- REQ-06: The player must receive feedback (e.g., sound effect) upon jumping.

3.3 Crouching Mechanic

3.3.1 **Description:** The player can crouch under certain obstacles. This feature is of **Medium** priority.

3.3.2 Stimulus/Response Sequences:

- **Stimulus:** The player presses the crouch button.
- **Response:** The player character crouches to avoid obstacles.

3.3.3 Functional Requirements:

- REQ-07: The player must be able to crouch and remain in that position until the crouch button is released.
- REQ-08: The crouch action must allow the player to pass under specific obstacles.
- REQ-09: The player must receive feedback (e.g., visual animation) upon crouching.

3.4 Dashing Mechanic

3.4.1 **Description:** The player can dash to eliminate enemies. This feature is of **High** priority.

3.4.2 Stimulus/Response Sequences:

- **Stimulus:** The player presses the dash button at the right moment.
- **Response:** The player character dashes forward, eliminating any enemies in the path.

3.4.3 Functional Requirements:

- REQ-10: The player must be able to perform a dash action that temporarily increases speed.
- REQ-11: The dash action must eliminate enemies upon contact.
- REQ-12: The player must have a cooldown period after using the dash ability.

3.5 Drug Consumption Effects

3.5.1 Description:
The game simulates the effects of drug use, impacting the player's health. This feature is of **High** priority.

3.5.2 Stimulus/Response Sequences:

- **Stimulus:** The player character collides with a drug item.
- **Response:** The player's health decreases, and visual effects are applied.

3.5.3 Functional Requirements:

- REQ-13: The game must reduce the player's health by a specified amount upon drug consumption.
- REQ-14: The game must apply visual effects (e.g., screen darkening) to indicate the negative impact of drug consumption.
- REQ-15: The player must receive feedback (e.g., sound effect) upon consuming a drug.

3.6 Health Management

3.6.1 Description:
Players can collect medicines to restore health. This feature is of **High** priority.

3.6.2 Stimulus/Response Sequences:

- **Stimulus:** The player character collides with a medicine item.
- **Response:** The player's health increases, and visual feedback is provided.

3.6.3 Functional Requirements:

- REQ-16: The game must increase the player's health by a specified amount upon collecting a medicine item.
- REQ-17: The player must receive feedback (e.g., sound effect) upon collecting a medicine.
- REQ-18: The game must display the current health status on the screen.

3.7 Enemy Interaction

3.7.1 Description:
The player can defeat or dodge enemies. This feature is of **High** priority.

3.7.2 Stimulus/Response Sequences:

- **Stimulus:** The player encounters an enemy.
- **Response:** The player can either dash to defeat the enemy or jump to avoid it.

3.7.1 Functional Requirements:

- REQ-19: The game must spawn enemies that patrol a defined area.
- REQ-20: The player must receive feedback (e.g., score increase) upon defeating an enemy.
- REQ-21: The game must have a defined health value for each enemy, which decreases upon being hit by the player.
- REQ-22: If the player collides with an enemy without dashing, the player's health must decrease.

3.8 Obstacle Generation

3.8.1 Description:
Random obstacles will spawn during gameplay, requiring player interaction. This feature is of **Medium** priority.

3.8.2 Stimulus/Response Sequences:

- **Stimulus:** The game progresses, and obstacles are generated.
- **Response:** The player must navigate around or over the obstacles.

3.8.3 Functional Requirements:

- REQ-23: The game must randomly generate two types of obstacles: those that require jumping and those that require crouching.
- REQ-24: The game must provide visual cues for the type of obstacle (e.g., color coding or shape).
- REQ-25: The player must receive feedback (e.g., sound effect) upon successfully navigating an obstacle.

3.9 Power-Ups

3.9.1 Description:
The game includes time-based power-ups that enhance player abilities. This feature is of **Medium** priority.

3.9.2 Stimulus/Response Sequences:

- **Stimulus:** The player collects a power-up item.
- **Response:** The player gains a temporary ability (e.g., invincibility, speed boost).

3.9.3 Functional Requirements:

- REQ-26: The game must include various power-ups that provide temporary effects.
- REQ-27: The effects of power-ups must be clearly indicated to the player (e.g., visual effects, UI indicators).
- REQ-28: The game must have a timer for each power-up effect, displaying the remaining duration to the player.

3.10 Game Start and Game Over

3.10.1 Description:
The game must have clear start and end conditions. This feature is of **High** priority.

3.10.2 Stimulus/Response Sequences:

- **Stimulus:** The player selects "Start Game" from the main menu.
- **Response:** The game begins, and the player character starts moving.
- **Stimulus:** The player's health reaches zero.
- **Response:** The game displays a "Game Over" screen with options to restart or exit.

3.10.3 Functional Requirements:

- REQ-29: The game must display a main menu with options to start the game, view settings, and exit.
- REQ-30: The game must transition to a "Game Over" screen when the player's health reaches zero.
- REQ-31: The "Game Over" screen must display the final score and options to restart or exit the game.

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4. Data Requirements

This section describes various aspects of the data that the system will consume as inputs, process in some fashion, or create as outputs.

4.1 Logical Data Model

The logical data model for "Daddict: Side-Scroller Edition" includes the following entities and their relationships:

- **Player**
  - Attributes:
    - health: Integer (current health points)

        - score: Integer (current score)
        - position: Vector2 (current position in the game world)
        - powerUps: List of PowerUp (active power-ups)
- **Enemies**
    - Attributes:
        - type: String (type of enemy)
        - health: Integer (current health points)
        - position: Vector2 (current position in the game world)
        - patrolArea: Vector2 (defined area for enemy patrol)
- **Obstacles**
    - Attributes:
        - type: String (type of obstacle)
        - position: Vector2 (current position in the game world)
        - interactionType: String (jump or crouch)
- **Power-Ups**
    - Attributes:
        - type: String (type of power-up)
        - duration: Float (time in seconds the power-up lasts)
        - effect: String (description of the effect)
- **Medicines**
    - Attributes:
        - healthRestoration: Integer (amount of health restored)
        - position: Vector2 (current position in the game world)

Note: A basic visual (tentative) representation of this data model:

https://app.eraser.io/workspace/05QFen2wDGJR3gQVTR2F?origin=share

Some User Stories:

1. *As a player, I want to select my preferred language at the start of the game, so that I can enjoy the game in a language I understand.*
2. *As a player, I want to customize my character's appearance, so that I can express my individuality and make my character unique.*
3. *As a player, I want to receive notifications about new updates or events in the game, so that I can stay informed about new content and features.*
4. *As a player, I want to have the option to save my game progress at any point, so that I can return to the game later without losing my achievements.*

5. *As a player, I want to access a tutorial mode that teaches me the game mechanics, so that I can learn how to play effectively without feeling overwhelmed.*
6. *As a player, I want to see a leaderboard that displays the top scores of players worldwide, so that I can compare my performance and strive to improve.*
7. *As a player, I want to have the ability to report bugs or issues directly from the game, so that I can help improve the game experience for myself and others.*
8. *As a player, I want to experience dynamic weather effects in the game, so that the environment feels more immersive and engaging.*
9. *As a player, I want to unlock achievements for completing specific challenges, so that I can have additional goals to strive for while playing.*
10. *As a player, I want to have the option to adjust the difficulty level of the game, so that I can tailor the experience to my skill level and preferences.*

4.2 Data Dictionary

The data dictionary defines the composition of data structures and the meaning, data type, length, format, and allowed values for the data elements that make up those structures.

Data Element	Description	Data Type	Length	Format	Allowed Values
health	Current health points of the player	Integer	N/A	0 to 100	0-100
score	Current score of the player	Integer	N/A	Non-negative	0 or greater
position	Current position in the game world	Vector2	N/A	(x, y)	Any valid coordinate
powerUps	List of active power-ups	List	N/A	N/A	PowerUp objects
type (Enemy)	Type of enemy	String	20	Alphanumeric	"xyz", "xyz", etc.
health (Enemy)	Current health points of the enemy	Integer	N/A	0 to 100	0-100
patrolArea	Defined area for enemy patrol	Vector2	N/A	(x, y)	Any valid coordinate
type (Obstacle)	Type of obstacle	String	20	Alphanumeric	"xyz"

| interactionType | Required interaction for the obstacle| String | 20 | Alphanumeric | "Jump", "Crouch" |

| type (Power-Up) | Type of power-up | String | 20 | Alphanumeric | "Invincibility", "Speed" |

| duration(PowerUp) | Duration of the power-up effect | Float | N/A | Seconds | Positive float |

| healthRestoration | Amount of health restored by medicine | Integer | N/A | 0 to 100 | 0-100 |

4.3 Reports

The game will generate the following reports:

1. **Player Statistics Report**: This report will summarize player performance, including total score, total enemies defeated, and total health collected.
   - **Characteristics:**
     - **Content**: Player ID, total score, total enemies defeated, total health collected.
     - **Sort Sequence**: By Player ID.
     - **Totaling Levels**: Total score and total health collected.

2. **Game Session Report**: This report will provide details of each game session, including duration, final score, and health status at the end of the session.
   - Characteristics:
     - Content: Session ID, duration, final score, health
     - Sort Sequence: By Session ID.
     - Totaling Levels: Total score and health status.

4.4 Data Acquisition, Integrity, Retention, and Disposal

This section describes how data is acquired and maintained, as well as the policies for data integrity, retention, and disposal.

1. **Data Acquisition:**
   - Player data (e.g., health, score) will be collected during gameplay and stored in memory. Upon game completion, this data will be saved to a persistent storage solution (e.g., local storage for mobile or a database for online play).
2. **Data Integrity:**
   - The system will implement validation checks to ensure that data is accurate and consistent. For example, health values must not exceed the maximum allowed value, and scores must be non-negative.

- Regular backups will be performed to prevent data loss, especially for player statistics and game session reports.

3. **Data Retention:**
    - Player data will be retained for a minimum of 30 days after the last game session. After this period, inactive accounts may be purged to free up storage space, provided that users are notified in advance.
    - Game session reports will be retained for a minimum of 6 months for analysis and improvement of game mechanics.
4. **Data Disposal:**
    - When data is no longer needed, it will be securely deleted to prevent unauthorized access. This includes removing player data from the database and clearing local storage on mobile devices.
    - Any residual data (e.g., cached data) will be managed according to the system's data retention policy, ensuring that it is not retained longer than necessary.

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5. External Interface Requirements

This section provides information to ensure that the system will communicate properly with users and with external hardware or software elements.

5.1 User Interfaces

The user interface (UI) for "Daddict: Side-Scroller Edition" will be designed to provide an intuitive and engaging experience for players. The following logical characteristics will define the user interfaces:

1. **Main Menu:**
    - Options: Start Game, Settings, Leaderboards, Exit.
    - Layout: Centered buttons with clear labels.
    - Standard Buttons: Each button will have hover effects and click animations.
    - Keyboard Shortcuts:
        - "Enter" to select an option.
        - "Esc" to exit the menu.
2. **In-Game UI:**
    - Health Bar: Displayed at the top left, indicating the player's current health.
    - Score Display: Located at the top center, showing the current score.

- Enemies Defeated: Text displaying the number of enemies defeated successfully.
- Power-Up Indicator: Shows active power-ups and their remaining duration.
- Pause Button: Accessible via the "P" key, bringing up a pause menu with options to resume, restart, or exit.

3. **Game Over Screen:**
   - Content: Displays final score, options to restart or return to the main menu.
   - Layout: Clear and prominent buttons for user actions.
4. **Error Messages:**
   - Format: Standardized error messages will appear in a pop-up box with a red border.
   - Example: "Error: Unable to connect to the server. Please try again."

5.2 Software Interfaces

"Daddict: Side-Scroller Edition" will interface with several software components, including:

- **Game Engine: Unity 6**
  - **Purpose:** The core engine for rendering graphics, handling physics, and managing game logic.
  - **Data Exchange:**
    - **Input**: Player actions (keyboard, mouse, touch).
    - **Output**: Game state updates (health, score, enemy positions).
- **Database: Firebase or a similar cloud database**
  - **Purpose:** To store player statistics, game sessions, and leaderboards.
  - **Data Exchange:**
    - **Input:** Player scores, health data, and game session details.
    - **Output**: Retrieved player statistics and leaderboard data.
- **Social Media APIs: Facebook, Twitter**
  - **Purpose**: To enable players to share scores and achievements.
  - **Data Exchange:**
    - Input: Player achievements and scores.
    - Output: Confirmation of successful posts.

***Nonfunctional Requirements*:**

- **Response Time:** All API calls should respond within 2 seconds.
- **Security:** All data exchanged with external services must be encrypted using HTTPS.

5.3 Hardware Interfaces

"Daddict: Side-Scroller Edition" will support various hardware interfaces, including:

1. **Supported Device Types:**
   - PCs (Windows and macOS)
   - Mobile Devices (iOS and Android)
2. **Input Devices:**
   - Keyboard: Standard keys for gameplay (WASD for movement, Space for jump).
   - Mouse: For menu navigation and in-game actions.
   - Touchscreen: For mobile devices, supporting tap and swipe gestures.
3. **Output Devices:**
   - Monitors: Support for various resolutions (minimum 1280x720).
   - Speakers/Headphones: For audio feedback and sound effects.

Timing Issues: The game must maintain a suitable frame rate of at least 30 FPS on all supported devices to ensure smooth gameplay.

5.4 Communications Interfaces

The game will utilize several communication functions, including:

1. **Network Protocols:**
   - HTTP/HTTPS for communication with the cloud database and social media APIs.
2. **Message Formatting:**
   - JSON will be used for data exchange between the game and external services.
3. **Security:**
   - All communications with external services must be secured using SSL/TLS encryption.
   - User authentication will be required for accessing social media sharing features.
4. **Data Transfer Rates:**

- The game should handle data transfer rates of up to 1 MB/s for leaderboard updates and player statistics.

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6. Quality Attributes

This section outlines the quality attributes that will define the performance and usability of "Daddict: Side-Scroller Edition."

6.1 Usability

To ensure that "Daddict: Side-Scroller Edition" is user-friendly, the following usability requirements will be implemented:

1. **Ease of Use:** The game will feature an intuitive interface that allows players to easily navigate menus and understand gameplay mechanics without extensive tutorials.
2. **Ease of Learning:** New players will be introduced to game mechanics gradually through a tutorial level that explains controls, objectives, and interactions.
3. **Memorability:** Key controls and gameplay mechanics will be consistent throughout the game, allowing players to easily remember how to play after breaks.
4. **Error Handling and Recovery:** The game will provide clear error messages and allow players to recover from mistakes (e.g., respawning after losing health).
5. **Efficiency of Interactions:** Common actions (e.g., jumping, dashing) will be mapped to easily accessible buttons to minimize player effort.
6. **Accessibility:** The game will include options for colorblind modes and adjustable text sizes to accommodate players with different needs.
7. **Ergonomics:** The UI will be designed to minimize strain on players, with appropriate spacing and sizing of buttons and text.

User Interface Design Standards**:** The game will adhere to established UI/UX design principles, ensuring a consistent and engaging experience across all platforms.

6.2 Performance

The following performance requirements will be established for "Daddict: Side-Scroller Edition":

1. **Frame Rate:** The game must maintain a minimum frame rate of 30 frames per second (FPS) on all supported devices to ensure smooth gameplay.
2. **Load Times:** Game should load within 5 seconds on average to minimize player wait times.
3. **Response Time**: User inputs (e.g., button presses) must be registered within 100 milliseconds to ensure responsive gameplay.

6.3 Security

The following security requirements will be implemented to protect user data and ensure privacy:

1. **Data Encryption:** All sensitive data exchanged between the game and external services (e.g., player statistics, social media sharing) must be encrypted using SSL/TLS protocols.
2. **User Authentication:** Players must create accounts and log in to access online features, with password policies.
3. **Data Privacy Compliance:** The game will comply with relevant data protection regulations to ensure user data is handled responsibly.
4. **Access Control**: Only authorized personnel will have access to the game's backend systems and databases.

6.4 Safety

To ensure player safety while using "Daddict: Side-Scroller Edition," the following requirements will be established:

1. **Content Warnings:** The game will include warnings about sensitive content related to drug use and addiction, allowing players to make informed choices about their gameplay experience.
2. **In-Game Safeguards:** Players will have the option to enable parental controls to restrict access to certain game features or content.
3. **Emergency Exit:** The game will provide a quick exit option to allow players to leave the game immediately if they feel uncomfortable.

***Safety Certifications*:** The game will adhere to industry standards for digital content, ensuring that it is suitable for the intended audience.

6.5 Additional Quality Attributes

1. **Availability:** The game will be available 99.9% of the time, ensuring minimal downtime for players.
2. **Reliability:** The game must not crash or exhibit critical bugs during normal gameplay, with a target of less than 1% of players experiencing crashes.
3. **Scalability:** The game architecture will be designed to support future updates and additional content without significant rework.
4. **Modifiability:** The game code will be structured to allow for easy updates and modifications, enabling the addition of new features and content.

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7. Internationalization and Localization Requirements

To ensure that "Daddict: Side-Scroller Edition" is suitable for use in various nations, cultures, and geographic locations, the following internationalization and localization requirements will be implemented:

1. **Language Support:** The game will support multiple languages, and will not be limited to English, Spanish, French, German, and Mandarin. Language selection will be available in the settings menu.
2. **Date and Time Formatting:** The game will display dates and times according to the user's locale settings. For example, the format will be MM/DD/YYYY in the U.S. and DD/MM/YYYY in the U.K.
3. **Currency Formatting:** If the game includes any in-game purchases or currency, it will display amounts in the local currency format based on the user's region.
4. **Cultural Sensitivity**: The game content will be reviewed to ensure it is culturally appropriate for different regions, avoiding any symbols, language, or themes that may be offensive or misunderstood in certain cultures.
5. **Character Sets:** The game will support Unicode to accommodate various character sets, ensuring that all languages can be displayed correctly.
6. **Time Zones:** The game will adjust any time-based events or features according to the player's local time zone.

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8. Other Requirements

This section outlines additional requirements that are pertinent to the project:

1. **Legal and Regulatory Compliance:** The game will adhere to all relevant laws and regulations, including those related to digital content, data protection (e.g., GDPR, CCPA), and age ratings.
2. **Installation Requirements:** The game will be available for download via major platforms (e.g., Steam, App Store, Google Play) and will not require complex installation procedures. A simple installation wizard will guide users through the setup process.
3. **Configuration and Startup:** The game will allow users to configure settings (e.g., graphics, audio, controls) before starting the game for the first time.

4. **Logging and Monitoring:** The game will implement logging for critical events (e.g., errors, crashes) to facilitate troubleshooting and improve future versions. Logs will be stored locally and can be sent to the development team upon user consent.
5. **Audit Trail:** The game will maintain an audit trail of user actions related to account management and in-game purchases to ensure transparency and accountability.

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9. Glossary

- **2.5D:** A term used to describe a visual style that combines 2D graphics with 3D elements, providing depth while maintaining a primarily 2D gameplay experience.
- **API:** Application Programming Interface; a set of rules and protocols for building and interacting with software applications.
- **GDPR:** General Data Protection Regulation; a regulation in EU law on data protection and privacy.
- **CCPA:** California Consumer Privacy Act; a state statute intended to enhance privacy rights and consumer protection for residents of California, USA.
- **UI:** User Interface; the means by which a user interacts with a computer, software, or application.
- **UX:** User Experience; the overall experience a user has when interacting with a product, particularly in terms of how enjoyable or efficient it is.
- **Vector2:** A data structure used to represent a point in 2D space, typically defined by two coordinates (x, y).
